

Suspensions & Tyres – Sharan Kalamohan (2518242)

1) Introduction

The reference vehicle for this study is the Tatuus F4 T-421, operating under the 2026 FIA Formula 4 technical regulations. These regulations define a controlled platform in which performance optimisation is allowed within a fixed architecture. The chassis features a 2750 mm wheelbase, with front and rear tracks of 1510 mm and 1460 mm respectively. Suspension is configured as a pushrod-actuated double wishbone system with rocker arms, coil springs, anti-roll bars, and two-way adjustable monotube hydraulic dampers at both axles. The regulations impose limitations on design freedom where only conventional coil springs and passive hydraulic dampers are permitted, with damping characteristics restricted to variation as a function of piston velocity. Active suspension, third elements, and interconnected or electronically controlled systems are prohibited. Furthermore, homologation fixes the structural design of key components, including wishbones, uprights, and kinematic geometry, preventing any changes to the design. Consequently, performance gains must be achieved through optimisation of the available setup parameters.

2) Literature Review

I. Overview

The suspension–tyre system governs vehicle–track interaction, with performance dependent on maintaining consistent tyre contact under dynamic loading. Tyre force generation is highly sensitive to vertical load, contact patch distribution, and operating state, requiring the suspension to control load transfer and wheel motion during both transient and steady-state conditions [9]. While advanced systems can independently control ride, roll, and pitch, these are prohibited by FIA regulations, constraining the system to passive operation [10]. Within this framework, the T-421 allows optimisation through setup parameters including ride height, camber, toe, anti-roll bar stiffness, and damping, which are all strongly coupled and directly influence tyre behaviour [11]. Tyre characteristics further govern system response, as vertical load, pressure, and camber affect deflection and effective stiffness [12]. Consequently, suspension and tyre behaviour must be treated as a coupled system, where performance is achieved through controlled load distribution rather than independent subsystem optimisation.

II. Suspensions

Racing suspension development focuses on minimising tyre load variation while maintaining platform stability. Advanced technologies aim to increase control over this interaction, though their implementation introduces trade-offs in complexity, cost, and regulatory compliance.

Active suspension uses actuator-generated forces to directly control ride height and suppress pitch and roll, maintaining optimal tyre loading under varying aerodynamic and road inputs [13]. While this provides superior performance, it requires high system complexity, external energy input, and advanced control architecture, making it both costly and prohibited under FIA regulations [13]. Semi-active suspension systems control damping characteristics in real time, typically using magneto-rheological fluids, improving transient response and oscillation control without requiring full actuation [14]. Although less complex than active systems, they still rely on electronic control and variable damping beyond piston-velocity dependence, making them incompatible with F4 regulations [14].

Heave spring systems decouple vertical and roll motion, enabling improved aerodynamic control and more consistent tyre loading [15]. Despite performance benefits, increased mechanical complexity, packaging constraints, and the prohibited use of third elements prevent their use in Formula 4 [15]. Geometry optimisation improves performance through control of camber gain, bump steer, and roll centre behaviour, enhancing tyre load distribution without additional hardware complexity [16]. Although full geometric redesign is restricted by homologation, these principles remain applicable through setup adjustments that influence tyre behaviour and vehicle response [16].

Overall, modern suspension development seeks increased control over tyre loading through actuation, adaptive damping, and geometric refinement. However, regulatory constraints in Formula 4 limit implementation to passive systems, requiring performance gains to be achieved through optimisation of homologated parameters [13]–[16]. The comparison of these technologies is shown in Table 10:

Table 10: Comparison table of suspension technologies

Suspension Technology	Performance Impact	Complexity / Cost	Reliability	F4 Legality
Active	Direct control of ride, roll, pitch to maximise tyre load consistency	Very High (actuators, control systems)	Low (system dependency)	No – external energy & active control prohibited
Semi-Active	Improved transient response & load control via variable damping	High (sensors and control)	Medium	No – damping must depend only on piston velocity
Heave Element	Decouples heave/roll to improve aero platform stability	High (packaging and tuning complexity)	Medium	No – third element explicitly forbidden
Kinematic Optimisation	Improves load distribution via camber gain, roll centre control	Medium (design and analysis)	High	Partial – geometry fixed, but setup adjustments allowed

III. Tyres

Tyres define the operating conditions within which suspension tuning is effective, acting as the force-generating interface between the vehicle and the track. Unlike rigid structural components, racing tyres show nonlinear behaviour due to its material properties, pressure, and temperature.

Tyre compounds determine friction generation and energy dissipation, requiring a balance between grip, wear, and rolling resistance [17]. This behaviour arises from viscoelastic properties of rubber, where composition and glass transition temperature influence energy loss and deformation characteristics. Advanced silica–silane filler systems improve performance trade-offs but increase manufacturing complexity and cost [17]. In Formula 4, compound selection is fixed, shifting performance focus to maintaining optimal operating conditions rather than material design. Tyre pressure directly influences vertical stiffness, deformation, and contact patch characteristics, making it a critical control parameter [18]. Higher pressures reduce deformation and rolling resistance but can reduce grip, while lower pressures increase compliance at the cost of stability. Although advanced systems can dynamically adjust pressure, these require additional hardware and control systems, making them both impractical and illegal in F4.

Thermal behaviour further constrains performance, as tyre friction peaks within a narrow temperature range and degrades outside this window [19]. Heat generation and dissipation occur through frictional, conductive, convective, and radiative mechanisms, resulting in complex temperature gradients within the tyre structure [19]. Load distribution directly influences thermal behaviour, linking suspension setup to tyre performance. Smart tyre technologies integrate sensing and estimation systems to monitor pressure, temperature, deformation, and friction-related parameters in real time [20]. These systems improve understanding of tyre–road interaction and enable advanced control strategies, but introduce challenges in durability, power supply, and integration. Their reliance on embedded electronics and data systems makes them incompatible with Formula 4 regulations [20].

Overall, tyre performance is governed by material behaviour, stiffness control, and thermal management. In Formula 4, where compound selection and advanced technologies are restricted, these effects must be managed indirectly through suspension setup, reinforcing the need to treat the tyre–suspension system as a coupled, constraint-driven optimisation problem [17]–[20]. The comparison of these technologies is shown in Table 11:

Table 11: Comparison table of tyre technologies

Tyre Technology	Performance Impact	Complexity / Cost	Reliability	F4 Legality
Advanced Compounds	Improves grip, wear, and rolling resistance via material optimisation	High (material + manufacturing)	Medium	Partial – compound fixed by supplier, no design freedom
Pressure Management	Alters stiffness, contact patch, and grip characteristics	High (sensors + pneumatic systems)	High	No – active pressure systems and control valves prohibited
Thermal Management	Maintains optimal temperature window for peak grip and wear control	Medium (monitoring + operational control)	Medium	Yes – passive management allowed, no active systems
Smart Tyres	Real-time measurement of pressure, temperature, deformation, friction	Very High (sensors + integration)	Low	No – embedded sensing and data systems not permitted

3) Numerical Setup For Suspension Optimisation

I. Input Parameters

The suspension numerical study was developed using the idealised spring–damper model from the suspension labs. The vehicle was defined with a total mass of 570 kg including the driver [21], a 40:60 front–rear weight distribution [22], assumed unsprung front and rear masses of 22 kg and 24 kg per corner respectively and a front and rear track of 1.51m and 1.46m respectively. A damping ratio of 0.675 was taken as common ranges for race cars are between 0.65 and 0.7. Suspension inputs were taken from the T421 manual and include discrete homologated spring options of 700–1100 lb/in with their respective vertical stiffness values [11]. Motion ratios for the front and rear were given as 1.1 and 1.23 respectively [11]. The tyre model was based on Pirelli F4 dry slicks with front and rear vertical stiffness values of 144,780 N/m and 175,180 N/m respectively, taken at a representative operating condition (1.4 bar, 0° camber). A local linear approximation was used [12].

The corner sprung masses can be calculated through Equation 1 and 2:

$$m_{s_{corner,f}} = \frac{0.4 \times m_{vehicle}}{2} - 22kg \quad \text{Equation 1}$$

$$m_{s_{corner,r}} = \frac{0.6 \times m_{vehicle}}{2} - 24kg \quad \text{Equation 2}$$

The ride rates can be calculated using vertical stiffness values through Equation 3:

$$K_R = \left(\frac{1}{K_W} + \frac{1}{K_T} \right)^{-1} \quad \text{Equation 3}$$

The natural frequency for sprung mass can be calculated through Equation 4:

$$f_s = \frac{1}{2\pi} \sqrt{\frac{K_R}{m_s}} \quad \text{Equation 4}$$

The critical damping and damping coefficients can be found, through Equation 5 and 6:

$$C_{crit} = 4\pi m_s f_s \quad \text{Equation 5}$$

$$C = 0.675 C_{crit} \quad \text{Equation 6}$$

The pitch ratio can be calculated through Equation 7:

$$\text{Pitch Ratio} = \frac{f_{front}}{f_{rear}} \quad \text{Equation 7}$$

Roll stiffness from each axle and front roll share can be found, through Equation 8 and 9:

$$K_\phi = \frac{K_w \cdot t^2}{2} \quad \text{Equation 8}$$

$$\text{Front Roll Share} = \frac{K_{\phi_f}}{K_{\phi_f} + K_{\phi_r}} \quad \text{Equation 9}$$

A parametric MATLAB study was used to evaluate the suspension setup where the script systematically swept all homologated spring combinations, recalculating ride rate, natural frequency, and damping requirements for each case. A weighted penalty-based optimisation method was used to evaluate each configuration against Silverstone-specific targets. For each parameter, a penalty function was defined, where changes from the target value is penalised. The overall performance score was then calculated. This approach ensures a balanced ranking across all configurations, capturing the trade-offs present in multi-parameter optimisation. The target values and weightings were selected to reflect the demands of Silverstone under dry conditions, prioritising sufficient front platform support while avoiding excessive stiffness that would increase pitch separation, a slightly softer rear axle to maintain traction and compliance, a pitch ratio greater than unity for longitudinal stability, a moderate front roll stiffness bias to control lateral load transfer, and unsprung frequency ranges that maintain consistent tyre contact.

4) Results & Discussion

All homologated front–rear spring combinations were evaluated using the developed suspension–tyre model and ranked using a weighted optimisation method tailored to Silverstone dry conditions. The ranking incorporated vertical stiffness, sprung ride frequencies,

pitch ratio, front roll stiffness share, and damping requirements. The complete ranking is shown in Figure 8:

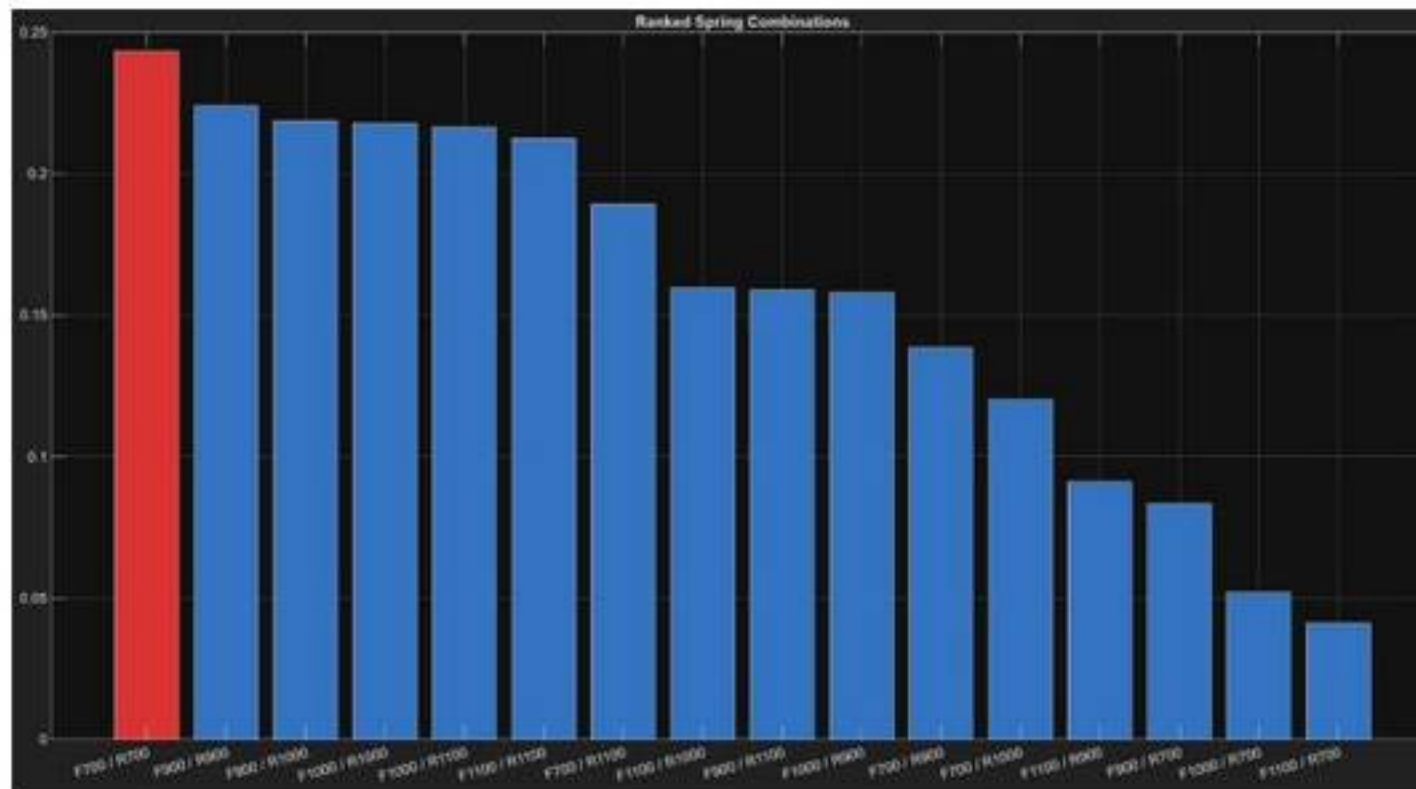


Figure 8: Overall ranking of all spring combinations using the weighted optimisation score.

The highest-ranked configuration was F700 / R700, with an overall score of 0.243. This setup produced front and rear ride rates of 59.60 N/mm and 55.40 N/mm, respectively, corresponding to front and rear sprung natural frequencies of 4.051 Hz and 3.090 Hz respectively. The resulting pitch ratio was 1.311, while the front roll stiffness share was 0.572, indicating a front-dominant balance. The corresponding damping coefficients required to achieve the target damping ratio were 3161.3 Ns/m at the front and 3852.6 Ns/m at the rear.

The calculated ride rates and frequencies are shown in Figure 9 which shows the variation in vertical stiffness and ride dynamics across all spring combinations:

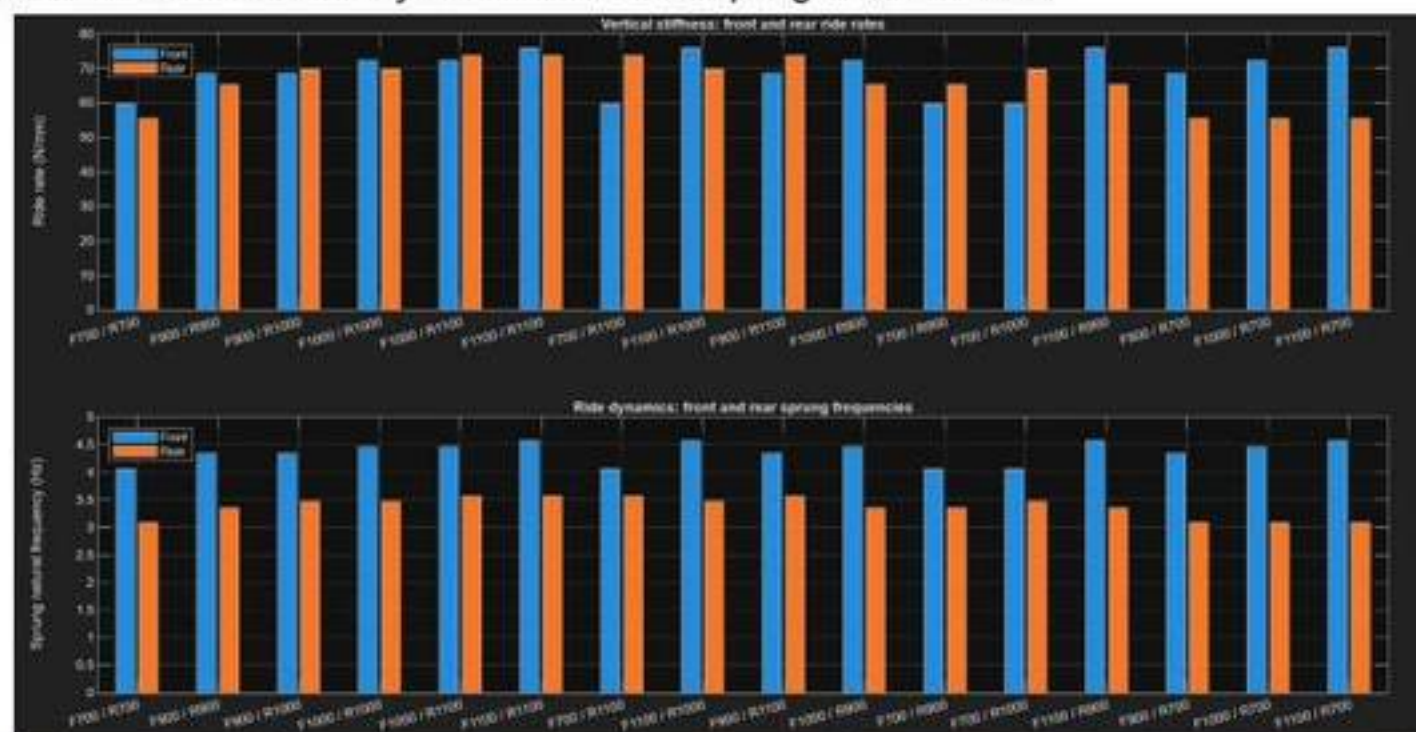


Figure 9: Comparison of front and rear ride rates and sprung natural frequencies for all evaluated spring combinations.

As expected, increasing spring stiffness increased both ride rate and sprung natural frequency at each axle. The front axle consistently showed higher frequencies than the rear due to the selected weight distribution, motion ratios, and stiffness levels. Configurations such as F900/R900, F900/R1000, F1000/R1000, and F1000/R1100 formed the next highest-ranked group, with scores between 0.216 and 0.224, indicating that several combinations provided similar overall performance trends. However, stiffer front-biased combinations such as F1000/R700 and F1100/R700 produced significantly larger pitch ratios and front roll shares, reducing their suitability for the selected optimisation targets.

The handling balance results are shown in Figure 10:

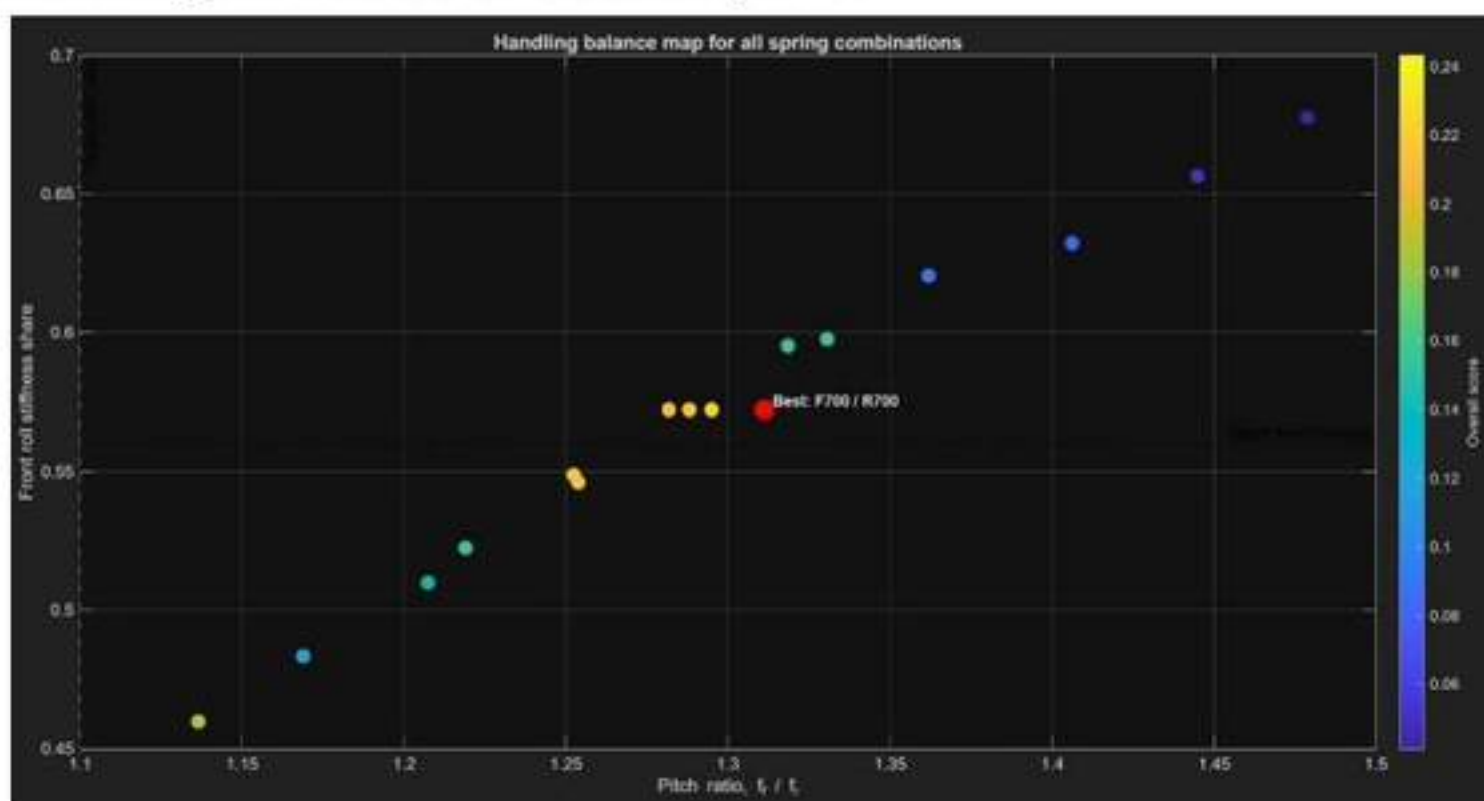


Figure 10: Handling balance map for all spring combinations based on pitch ratio and front roll stiffness share.

The selected setup combined a moderate pitch ratio with a front roll stiffness share close to the target region adopted for Silverstone. This suggests that the configuration provides a favourable compromise between front platform support and rear compliance within the simplified handling model, without introducing the excessive front stiffness bias observed in lower-ranked setups. By contrast, combinations with much stiffer front springs and relatively soft rear springs shifted strongly toward higher pitch ratios and larger front roll stiffness shares, indicating a more pronounced understeer tendency and poorer balance. Setups with much stiffer rear springs reduced pitch ratio but shifted the roll stiffness balance rearward, reducing their overall ranking.

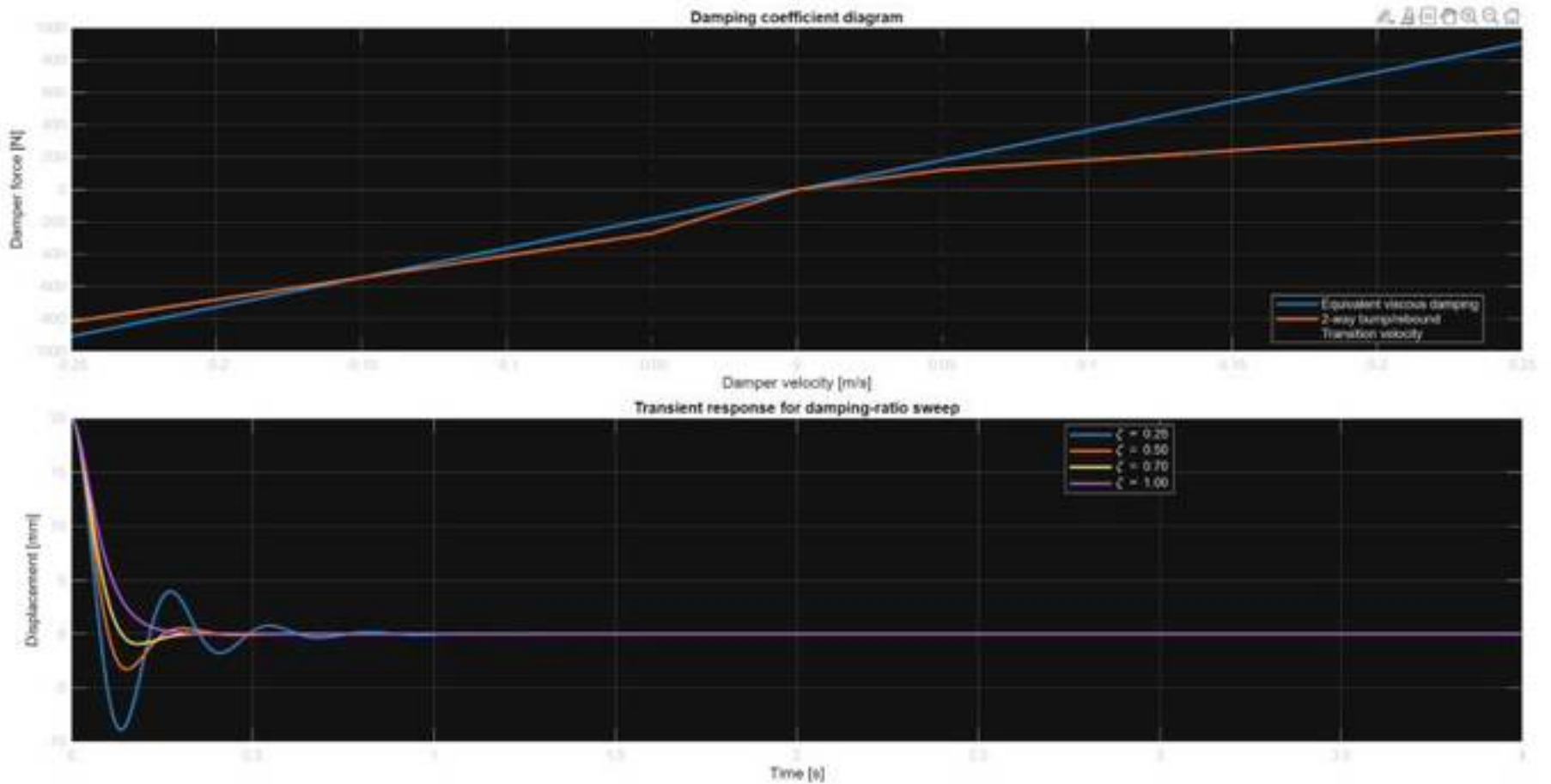


Figure 11: Damping optimisation summary for the selected optimum spring combination.

Figure 11 demonstrates both the damper force–velocity characteristics and the dynamic response of the suspension system under different damping ratios. The top plot shows the damping coefficient diagram, comparing an ideal linear viscous damper with a more realistic two-way damper model that separates bump and rebound behaviour and introduces a transition between low-speed and high-speed damping regions. This illustrates how damping can be tuned independently of spring stiffness to control both body motion and tyre contact. The bottom plot shows the transient response of the system to an initial displacement for different damping ratios, highlighting the trade-off between oscillation and response speed. Low damping results in significant oscillations and poor control, while critical damping eliminates oscillations but leads to slower return to equilibrium. The intermediate case provides the best compromise, minimising oscillations while maintaining a fast settling time, which is characteristic of well-tuned racing suspension systems.

In the absence of published setup data for the reference vehicle, a representative baseline configuration was defined using mid-range homologated spring options. The F1000/R1000 configuration was selected as it lies centrally within the available design space and provides a balanced distribution of front and rear stiffness without introducing extreme bias. The comparison between the baseline and the optimised setup is shown in Table 12:

Table 12: Comparison of Baseline and Optimised Spring Setup

Parameter	Baseline (F1000/R1000)	Optimal (F700/R700)	Change	Interpretation
Overall Score	0.218	0.243	↑ +0.025	Improved overall optimisation performance
Front Ride Rate (N/mm)	72.38	59.60	↓ -17.7%	Increased front compliance
Rear Ride Rate (N/mm)	69.70	55.40	↓ -20.5%	Increased rear compliance
Front Frequency (Hz)	4.464	4.051	↓ -9.3%	Softer front platform response
Rear Frequency (Hz)	3.466	3.090	↓ -10.9%	Improved rear compliance
Pitch Ratio (F/R)	1.288	1.311	↑ +1.8%	Increased front–rear separation
Front Roll Stiffness Share	0.572	0.572	No Change	Maintained lateral balance
Front Damping (Ns/m)	3412.0	3161.3	↓ -7.3%	Reduced damping requirement
Rear Damping (Ns/m)	4010.6	3852.6	↓ -3.9%	Reduced damping requirement

Compared to the baseline configuration, the selected optimal setup reduces both front and rear ride rates, resulting in lower overall vertical stiffness and increased compliance. The front sprung natural frequency decreases from 4.464 Hz to 4.051 Hz, while the rear decreases from 3.466 Hz to 3.090 Hz, maintaining a front-dominant frequency distribution. The calculated natural frequencies are higher than typical real-world F4 values due to the use of linearised tyre stiffness. The model is therefore used comparatively across homologated configurations rather than as an absolute predictor of vehicle behaviour. The pitch ratio increases slightly from 1.288 to 1.311, indicating a stronger separation between front and rear dynamic response. The front roll stiffness share remains the same at 0.572, maintaining a stable lateral load transfer distribution. This demonstrates that the optimal configuration improves compliance and dynamic balance while preserving key stability characteristics required for high-speed operation at Silverstone.

Steering & Brakes – Nithin Karehalli Ramesh

1. Introduction

The steering and braking subsystem controls how the vehicle changes direction and how it slows down. These two systems work together to allow the driver to handle the car safely and